SOLPS-ITER v.3.2.1 Release Notes

April 27th, 2026

These Release Notes provide a summary description of the code changes made since the launch of the Wide Grids 3.2.0 code version at the Leuven workshop in October 2022. A lot of this work was done under contract by the KUL team and detailed reports are provided as the contract deliverables (see links below). The material contained in these reports will not be repeated here. Thus, please refer to:

- ITER_D_A78A4Y : A detailed description of the improved **DivGeo-Carre-Triang** workflow to build WG cases
- ITER_D_CWAZY9 : Final contract deliverable for the Wide Grids updates contract by KUL
- ITER_D_CW9XZ6 : Report on implementation of Crank-Nicolson time-stepping scheme and improvements to MMS code validation tools

A large part of the changes in this update mirror similar ones done in the 3.0.10 structured code version, so users should familiarize themselves with the 3.0.10 (and previous versions) Release Notes to get a feel for what has been done there. Users should also consult the updated code inline documentation and extensively rewritten dedicated Wide Grids SOLPS-ITER manual that accompanies this distribution and other documents posted to the SOLPS-ITER web site and Confluence page. In particular, the minutes from the SOLPS-ITER User Forums will provide more detailed information about the features discussed below. For full details of all updates, please consult the GIT repository logs.

This new release is also the first one of the Wide Grids code version that will be made open-source on https://github.com/iterorganization/SOLPS-ITER, and contains some changes that are the consequences of this change of hosting, including updated licence files and code download instructions.

How to obtain SOLPS-ITER v.3.2.1:

Because of the change of hosting from the ITER GIT server to GitHub, it is recommended to clone a brand new tree in which to house the 3.2.1 code version. The initial clone will get the master version, after which the user should run the *solps_iter_update_wg_release* script. For later updates, the script can be invoked directly, but, before proceeding, please make sure that any local changes you might have are committed to local GIT branches or stashed, so as to avoid them being overwritten or creating conflicts. It is also recommended to have, as a separate side tree, a current 3.0.10 code version for reference. For users without access to the **AMNS ADAS** repository, which is not (yet) available on GitHub, a work-around is in place, by which the update script will detect that **AMNS ADAS** is unreachable and instead fetch a (large) selection of data files from the OpenADAS website and, for bundled datasets, a Zenodo record.

IMPORTANT: Because code updates may involve changes to configuration files, new environment modules (in particular, if you are using IMAS, it is recommended to link to IMAS/4.1.0 or younger, but at least version 3.37.0), and some number of added/renamed/moved/removed source code files, it is expected that the compilations launched by the update script will fail the first time around. When this happens, start afresh in a new window, load the SOLPS-ITER environment as usual by means of the *setup.csh* file, and try again to compile. If it fails once more, then do a "*gmake clean*" first and compile again. Recall that you can use "*gmake clean_XXX*" to remove all objects associated with any *XX* build, for example "*gmake clean_solps_mpi*".

N.B.: When restarting a 3.2.0 run with the 3.2.1 code version, the *b2time.nc* file contents may no longer match, so the time trace archival should be restarted (see below in output section).

Changes to the physics model:

Please be mindful that the changes to the default physics model described in this section will most likely cause some evolution of existing 3.2.0 solutions. The magnitude of those changes will depend on the specifics of the cases at hand. The benchmarks and sample cases available as part of the code distribution have all been recomputed using the 3.2.1 physics model.

- The prefactor multiplying the heat flux due to anomalous transport can now be toggled between 3/2 (new default) and 5/2 (old default) by means of the `keps_anom_he_model` switch.
- The Zhdanov-Grad closure is now available in the Wide Grids code, in much the same way as it is in the 3.0.10 structured code version. Neoclassical corrections to the thermal and friction forces applied to heavy impurities can be included by means of the `b2tral_Zhdanov_nc` switch.
- The `b2tqna_user_transport.eq.6` option has been revived. This allows for specification of transport coefficients for a H-mode plasma, i.e. with three transport levels for the deep core, pedestal region, and SOL, respectively (plus some additional refinement options, see the code documentation for full details).
- The transport model for boundary cells at the end of a flux tube where it is being "shaved" by the wall has been made to mimic a situation where the full set of cells in contact with the wall is treated as a single "super-cell". See usage of the `b2tfnb_avg_vExB`, `b2us_prep_mod_hc`, and `b2us_prep_set_beta_0` switches, as described in the code documentation.
- Flux-scaling of radial transport coefficients is now available.
- The neoclassical module for computing transport coefficients in the confined region has been ported to the Wide Grids code.
- For flux tubes where the number of collisions is small (very short field lines and/or low density conditions), it is possible to reduce the transport coefficients and adapt the flux limiters by means of the `b2trcl_min_collisions` switch. The amount by which the viscosity is reduced is given by the `b2tlmv_cvsa_low_col_mltpl` switch, with a smooth transition depending on the actual collisionality in the flux tube considered.
- New feedback schemes are available, on the ion density at the outer midplane (`FB_TYPE=17`), the total electron cooling power (`18`), the neutral pressure in the PFR (`20`), the total radiation losses (`21`), the average separatrix density of a species (`22`), the total plasma heat flux on the outer target (`23`), the peak electron temperature (`24`), density (`25`), or parallel saturation current (`26`) along a boundary, and the total particle content of a species outside the confined region (`27`). A new actuator `FB_ACTUATOR=8` can be used to act on the pumping albedo of **Eirene** surfaces. Rescaling schemes have been corrected to match their documentation description. A new `FB_RESCALE_OPTION=7` setting provides a *tanh* rescaling, option `8` is a SOLPS4.3-style rescaling and option `9` is as option `7` but multiplying the pumping rate. See the code for details.

- Boundary conditions types BCCON=4, 16, 19, 21, 22, 23, 25, 26, 27, BCMOM=4, 8, 9, 10, 12, BCENE=4, 5, 12, 13, 16 (style=1), 17, 18, 20, 21, BCENI=4, 5, 12, 13, 16 (style=1), 17, 18, 20, 23, 24, 25, 26, 27, and BCPOT=4, 5, 9, 16, 21, have been adapted to the Wide Grids code. The capability for feedback against a flux-surface-averaged quantity is however only available for the core boundary surface.
- The set of drift-compatible sheath boundary conditions (BCCON=14, BCMOM=13, BCENE/I/K=15, BCPOT=11) now uses, across all of them, the same MOMPAR(IS,IB,1) parameter to specify the requested Mach flow velocity and the CONPAR(IS,IB,1) parameter is no longer used. This is done to avoid users needing to specify the Mach velocity twice, and we keep the parameter that is more obviously connected to a velocity. Consistency of usage of these BCs together as a group is also now more strictly enforced.
- For boundary conditions that use an internal feedback scheme, the strength of the feedback can be modified by means of the b2stbc_bc_ref (for density BCs), b2stbc_bc_ref_te (for electron heat and potential BCs), and b2stbc_bc_ref_ti (for ion heat BCs) switches, conditioned on the value of the second parameter for the corresponding boundary condition (default values are 10^{-2}). See code for details.
- For sheath BCs, in quasi-tangent cells where the sine of incidence angle is less than Qalfmin, the sheath BC is applied as if the sine of the incidence angle was Qalfmax. It is also now possible to choose a non-marginal formulation of the sheath BCs, relevant for multi-fluid runs (see Manual for details).
- Usage of the b2tfnb_vis_q (multiplier to the ion flows due to heat viscosity) and b2tfnb_vis_per (multiplier to the ion flows due to perpendicular viscosity) switches has been made consistent, so that these switches act on both the poloidal and radial ion flow components.
- Various extensions of the code were made to enable the fluid-kinetic hybrid model and improve upon the Advanced Fluid Neutrals (AFN) models, and to refine the *k*- model.
- Radial profiles of drift multipliers for **E×B**, diamagnetic, and viscosity-related terms, respectively, are now available.
- The computation of the electron **E×B** velocity has been corrected to use face-centered values for the magnetic field, instead of interpolating from cell centres.
- The kinetic energy flux fhm formulation was modified to avoid negative energy fluxes flowing from the wall into the plasma.
- The species-specific rescaling scheme for particle balance in **Eirene** has been implemented.
- A bug fix was made to get the correct momentum source in **Eirene** for heavy-heavy collisions between species of different mass.
- The code can now support cases without any hydrogenic species.
- A new "DT" hydrogenic species with mass 2.5 is now supported for **B2.5** runs.
- The Algorithmic Differentiation code capabilities have been extended and essentially match all the changes indicated here. See the Manual for details.
- The COCOS number for the SOLPS-ITER coordinate system convention has been corrected from 6 to 4.
- The fundamental constants have been updated to the CODATA 2022 standard.

Changes to the input files:

IMPORTANT: Because the **Eirene** version coupled to the Wide Grids code is now the MsV version, users must make sure that their **Eirene** input file matches the MsV conventions. Please consult the **Eirene** manual and make the necessary changes. The automated translation tool **b2yt**, which can translate an **Eirene** input file from an older version to the MsV format, currently only supports structured grids, so, if you are converting a structured case to an unstructured one, you should do the **Eirene** input file conversion first with **b2yt** before running **b2us** for the **B2.5** input file conversions.

- The **Eirene** input file can now be provided in JSON format. If **Eirene** is run using the fixed form format, the code will then output a JSON version of the input file with the same contents (if the JSON library is available on the system, of course).
- A new **edit_geo** Matlab script is available to modify **Carre2** grids, with an associated Manual (automatically built by the *Makefile*) in the documentation folder.
- The *b2fgmtry* file now lists, at the end, the values of Qalfmin and Qalfmax used during the gridding process to adjust near-tangential boundary cells, making the b2stbc_Qalfmin switch obsolete, replaced with b2us_prep_Qalfmin, with the appropriate **b2yt** conversion rule. If they are not provided, a default value of 10^{-3} is used for both Qalfmin and Qalfmax. The magnetic field calculation in guard cells has been corrected and a new array geo%cvHy provides the effective radial width of control volumes. The poloidal magnetic field direction (either clockwise or counterclockwise around the plasma) is stored in geo%signmf.
- A new switch b2mndr_temp_ignore_geo can be used to skip the automatic detection of X-points and strike-points, for cases that do not match the standard tokamak topologies, and it is now possible to have a slab geometry with an artificial core region using the artificial_slab switch.
- The *ionization_potentials* file has been updated with the NIST 2024 data.
- A new adas_extrap switch is available to choose how to extrapolate ADAS rates beyond the range for which they are provided in the original data files. The default behaviour now limits the dependence on electron density and temperature to be not stronger than the ±3/2 power, as opposed to a simple log-log extrapolation (the previous default). Another available option is no extrapolation, using the nearest valid data point from the ADAS original file. See the code documentation for details. The check to make sure that the rtqr array does not go negative has also been corrected.
- The code can now handle cases in which the number of strata in the *fort.44* file read at the start of run is offset by one with the number required by the current run parameters, i.e. because the time census stratum is being added or removed.
- It is now possible to provide profiles of transport coefficients and/or sources by means of the *b2.transport.inputfile* and *b2.sources.inputfile* namelists, respectively.
- It is now possible to provide a list of control volumes, by means of the PFR_CVS array in *b2.user.parameters*, for computing neutral pressure.
- Conversion rules in **b2yt** and input sanity checks have been added for the *b2.feedback_control.parameters* file.
- A new program, **b2ab**, is available to build up tables of sputtering and reflection coefficients for AFN cases. Its usage is detailed, along with the derivations underlying the AFN model, in the files provided in the *$SOLPSTOP/doc/AFN_BCs_documentation* folder (PDF description automatically built by the *Makefile*).
- The largest AFN dataset files (*dbe.fnn*, *dc.bnn*, and *dw.fnn*) have been moved from the Git repository (they were too big for transfer to GitHub) to being distributed *via* a Zenodo record and are now automatically downloaded by the *Makefile* during code installation.
- The **Uinp** program now provides the BCENK settings in *b2.boundary.parameters.stencil*. It also provides a *b2mn.dat.stencil* skeleton file, relevant for vessel mode cases. By default, a time census stratum is no longer assumed in *b2.neutrals.parameters.stencil* nor in the **Eirene** input files (which are now provided in the MsV format), unless it is specifically requested, by setting NPRNLI to a non-zero value in the **DivGeo** Global Eirene data dialog box.
- If EIRENE_STEP_DT is not provided in *b2.neutrals.parameters*, the **Eirene** timestep is taken from the DTIMV variable in block 13 of the **Eirene** input file.
- The format for specifying reaction lines for line-of-sight diagnostics in **Eirene** (in block 12 of the input file) has been modernized, and now relies on existing reaction cards from block 4. Check the **Eirene** Manual for details.
- A target shaping factor for heat load calculations and parameters for computing the ballooning parameter can now be provided in *b2.user. parameters*.

- The switches `b2sikt_aniso` and `b2optim_reset_iter` have been renamed `b2sikt_fac_aniso` and `b2optim_reset_drift_iter`, respectively. A conversion rule in **b2yt** is provided.
- A new *setup_baserun_B25_links* script is available to make sure all necessary linked files from the *baserun* and database directories are present in the run directory for a **B2.5** standalone run. When using an **Eirene** input file in JSON format, use *setup_baserun_eirene_json_links* instead.
- In **Carre2**, two new parameters, `dpol1max` and `dpol2max`, are available to force removal of small triangular cells in the direction along or perpendicular to the flux surfaces, respectively. Both carry a default value of `0.01`.
- The *tiara_ggd_to_b2us.py* script allows for translation of a GGD object created by **TIARA** into a *b2fgmtry* file format.
- The `MAXPOIN` variable in *b2.neutrals.parameters* has been made obsolete, as the code now automatically sets up the size of the outer grid contour arrays.

Changes to the output and run post-processing tools:

- The **b2plot** code for post-run result visualization and manipulation is now available, with essentially all the same features as per the 3.0.10 structured code, and some more capabilities specifically related to unstructured code quantities. Some limited features are conditioned to having a magnetically rectangular base grid. Fluxes are interpolated from cell faces to cell centres so will not correspond exactly to those actually used in the code. In order for **b2plot** to access all the needed quantities, the *b2fplasma* file content has been expanded, mostly with arrays from the *srw* `B2SourceWork` data structure. In particular, wall loading files (including radiation) for the full contour including plasma fluxes for vessel mode cases, even for AFN runs, can now be produced. Please consult the SOLPS-ITER Manual for a full description.
- Saving simulation data (either during or after a run) as IMAS data entries is fully supported (up to DD/4.1.1 and AL/5.5.0, including backward-compatibility to AL4 and DDv3). For vessel mode cases, the `wall` IDS is also populated. When filling in GGD data, grid subsets corresponding to non-existent grid regions are skipped, but their index still matches the region index from the internal **B2.5** numbering.
- When an IMAS data entry is part of the run's output, the code will create a SimDB manifest file (*manifest.yaml*). This file is intended to contain a list of all the input and output files involved in the run, as well as a unique alias identifier and verbose description field. Users should check that the information the code wrote in the manifest is sufficient and complete, and make sure the description field is sufficiently explicit to uniquely identify the run and its purpose. Then, the file can be pushed to a SimDB database by means of the new *ingest_simulation_to_SimDB* script. This feature requires IMAS Access Layer 5.
- The *tracing* files allowing to follow the evolution of a run have been revived. Some new quantities have been added within *user.trc*, such as the peak parallel heat flux to the main chamber wall, ballooning parameters, and total saturation currents to the targets. Various examples of command files to plot the tracing data are now available in *$SOLPSTOP/scripts/commands*.
- The `Balance` Matlab scripts are working again, and the quantities from the *balance.nc* file are all available, in particular the sources due to neutrals computed by **Eirene**.
- New `netCDF` file output options are available, that will go in the *b2movies.nc* and *eirenemovies.nc* files. Consult the Manual for a full description.
- The **B2.5** and **Eirene** fluxes and anomalous transport coefficients fields on the **B2.5** grid as now available among the output in the *b2movies.nc* and *eirenemovies.nc* files, respectively.
- Several other quantities were added to *b2time.nc*, in particular with regards to multi-species quantities and **Eirene**-related quantities, such as densities for all fluid species (`nasep[a|i|m]`), atomic and molecular densities from **Eirene** (`d[a|m]bsep[a|i|m]`), atomic and molecular temperatures from **Eirene** (`t[a|m]bsep[a|i|m]`), density profiles for all fluid species (`na3d[a|i|l|r]`), atomic and molecular density profiles from **Eirene** (`d[a|m]b3d[a|i|l|r]`), atomic and molecular temperature profiles from **Eirene** (`t[a|m]b3d[a|i|l|r]`). The midplane transport coefficients profiles (`dn3d[a|i]`, `vx3d[a|i]`, `vy3d[a|i]`, `vs3d[a|i]`) and target particle flux profiles (`fn3d[a|i]`) are now given for all species, not only the main ion. This addition, because it requires adding new dimension definitions in the header of the *b2time.nc* file, is NOT backward-compatible and thus requires the user to restart the time trace archival when continuing a 3.2.0 run with the 3.2.1 code version (by removing/renaming the old *b2time.nc* file or setting `b2mndr_stim` to a zero or positive value).
- To have *b2time.nc* be fully self-contained, it now includes the radial profile coordinates (`dsi`, `dsa`, `dsr`, `dsl`), target areas (`dsR`, `dsL`) and poloidal contact areas (`dsRP`, `dsLP`), as well as the lists of control volumes and cell faces defining the midplanes and targets.
- To avoid issues on case-insensitive file systems where files like *dsr* and *dsR* would be identical, the latter are renamed to *dsRT*. Samd for *dsL* becoming *dsLT*.
- The *2d_plots* and *2d_plots_av* scripts have been expanded to include now also species-resolved time traces (plotted together according to the isonuclear sequence to which they belong). For consistency and clarity, the generated PostScript file has been renamed from *b2time2d(_av).ps* to *b2time2d_time_traces(_av).ps*.
- The *2d_profiles* script has been expanded so that it prints *.last10* (pro)files also for multi-species fields (namely one file for each fluid species). Since, for simulations with a large number of species, the number of produced files could become very large, potentially cluttering a lot the run directory, these files are now put in a *2d_profiles/* subfolder of the run directory. Additionally now, when the script is invoked, all profiles are also plotted to a new PostScript file, named *b2time2d_profiles.ps*.
- A new *summarize_run_tracing* script is available to obtain run analysis graphs based on data stored in the *tracing/* files, as opposed to *summarize_run* that uses the tallies written in the *run.log* output.
- **Eirene** now computes and provides vectorial momentum sources, along the parallel, radial, and diamagnetic directions. These are also passed on to **B2.5** and available in the *b2fplasma* output.
- Angular-resolved incident spectra tallies can now be obtained in **Eirene** by means of the `ISPOPT` and `ISPLDEG` switches in block 10 of its input file, in the form of Legendre polynomial expansions.
- A new `-DCHECKBIN` pragma is available to obtain ASCII copies of the binary files produced by **Eirene** to check their contents.
- The standard deviation tallies in **Eirene** are now available as standard output if requested.
- Sputtering tallies from **Eirene** can now be split according to the species that caused the sputtering, by means of the `NLSPCSPT` switch in block 1 of the input file.
- Saving archival run data using MDSplus on the IPP server with the **b2md** program (and reading it back with **b2rd**) is now again possible.
- Other post-processing programs have been revived: **b2yh**, **b2yr**, **b2yv**, **plasmastate_inspect**, and the conversion program **b2yt** (except for grid changes). The **Eirene** input file produced by **b2yt** will conform to the MsV format.
- Many new debugging output files can now be obtained with the `b2wdat_iout.eq.4` setting. Recall that use of this switch should only be done for short runs with debugging or detailed result analysis purposes, as they represent a very large I/O volume and can significantly affect the cluster's speed on which the runs are performed if used routinely. Use of this switch in production runs is indeed actually forbidden on the EUROfusion `ITM` Gateway and `PITAGORA` machines.
- Errors in the mapping of **Eirene** vectorial quantities (`[r|p|t]flux[a|m]`) to the **B2.5** coordinate system were fixed, as well as unit mistakes for the molecular source rates `srcml` and `edissml`.
- The `qconvereg` tally for convective electron heat flux has been corrected to take into account the parallel current. A faulty summing rule affecting the `radarr` tally was also corrected. Several other tallies have also been revived.
- When using one of the standard scripts for plotting residuals, those of the neutral temperature, turbulent **E×B** kinetic energy, and turbulent **E×B** enstrophy equations will also be included.

- New **b2plot** commands include *fteq* to select a particular flux tube, and *colle* and *colli* to obtain the average number of electron and ion collisions in a flux tube.
- The format of the numbers printed out in the wall loading tables produced by **b2plot** (with the *wlld* command) or in the *tracing* output files was changed so 10^{-100} will be written as `1.000E-100` instead of `1.000-100`, which could confuse scripts reading these numbers.
- When running **Carre2** with an extended equilibrium, this extended equilibrium will be written out in the *rzpsi_ext.dat* file.
- The main *Makefile* now has an *echo* target that allows to print out various internal variables for debugging.

Numerical improvements:

- The code has been modified to properly handle cases where the grid and plasma topologies differ from each other, and various bug fixes for CDN topologies implemented.
- Restart effects in coupled runs and some feedback schemes have been identified and fixed, and now systematically tested against as part of the CI procedure.
- Lots of input sanity checks have been added and formats corrected.
- The PARDISO (Parallel Direct Solver) from the Intel MKL library can now be used for solving the **B2.5** equations, by setting the `b2uxus_style` switch to 3 (instead of the default of 2 that uses the MA28 solver).
- The MMS validation tools have been extended.
- Various MPI (in **Eirene**) bottlenecks were identified and fixed to speed up parallelized cases. The `IMPROVED` MPI parallelization strategy has also been implemented.
- OpenMP parallelization is now available for **Eirene**, in addition to the previous **B2.5** implementation. Be aware that OpenMP parallelization in **Eirene** is done over the strata, while in **B2.5** it is done over the number of fluid species in the problem. These two numbers are different, so finding the optimum number of threads is case-specific. Also, it has been noticed that coupled OpenMP runs with a small number of threads can actually be SLOWER than an equivalent serial run. This behaviour is still under investigation.
- The **B2.5** plasma solver now has a full OpenMP parallelization implementation.
- New time-stepping schemes based on the Crank-Nicolson and Backward Differentiation 2 methods have been implemented, as well as cell-based timestep multipliers.
- A new set of switches is available to provide minima and maxima to the primary code variables, in both **B2.5** and **Eirene**.
- The new switch `b2nppo_restr_po` controls the maximum ratio between electron temperature and electric potential. This switch replaces `b2nppo_restrict_po`, with a **b2yt** conversion rule associated. Similarly, the switch `b2upht_rte_min` provides a way to impose a minimum ratio of electron to ion temperature.
- Modifiers to the time-stepping in the far SOL are now available in *b2.numerics.parameters*.
- As per the structured code, it is now possible to use the `nrings_for_no_speedup_averaging` switch to turn off the flux surface averaging speed-up scheme in the `nrings` outermost core region flux tubes.
- The **Eirene** source multiplier speed-up scheme has been implemented.
- The under-relaxation scheme for the **Eirene** sources has been revived and extended, in particular to avoid restart effects. These sources are saved in the *b2favere/i* files.
- When running **Eirene** in time-dependent mode, the neutral background is always checked for at the beginning of the run and created if not already present. The number of trajectories in the time census stratum is set to `NPRNLI` to avoid cascade overflows.
- The treatment of Fokker-Planck collisions in **Eirene** has been improved (as per the treatment in SOLEDGE3X). This is governed by the new `NLSOLEDGE` switch (recommended to be set to `.true.`, while `.false.` will recover old behaviour).
- New switches `NEVNTMAX` and `NVNMXTR` in **Eirene** (block 1 of the input file) allow for imposing an event number limit on individual trajectories to prevent them from being too long-lived.
- Switches to modify the flux limiters are now available, to apply in the far SOL and/or only for densities within a certain range.
- Heat sources in boundary cells can be turned off by means of the `b2sihs_shi*` switches. In the same cells, it is possible to increase the transport coefficients with the `b2txcx_increase_transp_coefs` switch, and modify their geometrical properties by means of the `b2us_prep_mod_hc` and `b2us_prep_set_beta_0` switches.
- New switches `b2stbc_stab_coeff_sheath_t[e|i]` are available to help stabilize the plasma temperatures where sheath BCs are applied.
- The scheme for stopping internal iterations or a full run when a certain low residual level has been achieved has been extended to also apply to momentum residuals.
- Guard cells are no longer considered to belong to a flux tube since they only neighbor their boundary cell.
- An upper limit has been added in **b2ar** to prevent atomic rates associated with energy losses to increase to too large unphysical values, when the *b2frates* data tables being built are being extrapolated beyond the range of validity of the original ADAS files.
- The maximum number of neighbours to a control volume has been increased to 20, and the maximum number of cell volume regions to 100.

Changes to the run environment:

Independently of this update, it is recommended to have installed v.1.2.5 of the MSCL library (available here), compiled with the same compiler as the SOLPS-ITER code components. You can use the new *install_dependencies* script to make sure you have the most recent version of the dependent libraries distributed by the IO.

- Compilation of **Eirene** is now handled, by default, via `CMake`. If you do not wish to use `Cmake`, you can set the environment variable `NO_CMAKE` to : to revert back to the traditional *Makefile* method, for which the *Makefile* has been moved to the *config/* directory to avoid a clash during build.
- The compilation configuration files have been streamlined to take maximum advantage of the *config.common.$COMPILER* files. It is now expected that the compiler variables `FC`, `CC`, and `MPI_FC`, used by the *Makefiles* to specify the Fortran serial, C serial, and Fortran MPI compilers, respectively, are provided in the *$SOLPSTOP/SETUP/setup.csh.$HOST_NAME.$COMPILER* files. Use of the *ifx* compiler, at least for the 2023 and 2024 APIs, is discouraged, as it is known to lead to fatal errors when running **Uinp** and OpenMP-parallelized simulations.
- When starting a new session, it is possible to override the default settings for the `HOST_NAME` and `COMPILER` variables by means of the environment variables `SOLPS_HOST_NAME_FORCE` and `SOLPS_COMPILER_FORCE`, respectively.
- The code supports both IMAS Access Layer 5 (recommended) and, for legacy cases, Access Layer 4.
- Configuration files have been provided for the new EUROFusion Gateway (EFGW) machine (using the old `ITM` hostname), the `NSCC` cluster in China, the `IPPCR` one in Prague, and the new `PITAGORA` machine, as well as for the `gfortran` toolchain at `PPPL`. The `IPP`, `LEUVEN`, and `ITER` files have been modified to match their respective new cluster installations.
- New configuration files have also been provided for standalone `LINUX` machines, MacOS `DARWIN`, and container environments.
- The job submission template files have been consolidated so that only one per cluster is needed as opposed to four previously. The job submission scripts now manage internally whether to compress the output and/or if the run is a **B2.5** standalone or **B2.5-Eirene** coupled one.

- The **Sonnet-light** library is no longer required for compilation of the SOLPS suite, unless one wishes to build the SOLPS4-to-5 **b2sxdr** converter.
- The example case files (*$SOLPSTOP/runs/examples*) are now distributed *via* a Zenodo record, which can be found as part of the SOLPS community resources. They can be downloaded as before using the dedicated *Makefile*.

Additional notes for developers:

1. To match the **Eirene** MsV naming convention, the interface routine folders *couple_SOLPS-ITER/* and *couple_SOLPS-ITER_structured/* have been renamed *coupled_SOLPS_WG/* and *coupled_SOLPS-ITER/* respectively.
2. The **Eirene** source files that require preprocessing have all been renamed to have extensions in *.F* or *.F90* instead of *.f* and *.f90*.
3. The Fortran standard applied to the code is FORTRAN 2003. Use of newer features is still allowed, provided it is surrounded by `#ifndef LEGACYCOMP` pragmas and, even better, if alternative coding is provided for older compilers. The `-F2003` pragma is no longer used.
4. To match the requirements of modern compilers, `common` blocks have been replaced by modules wherever possible and no new ones should be introduced in the code base.
5. The *DIMENSIONS.F* file that contained problem size definitions has been replaced with a *b2mod_dimensions.F* module. Where relevant, code is made backward compatible with a `-DDIMENSIONS_MODULE` pragma.
6. The *ioflush.c* routine from **Eirene** has been replaced by a user *ioflush_usr.F* routine.
7. The **Eirene** routines *eirene*, *sheath*, *timep*, *timer*, *upcusr*, and *uptusr* have been modularized.
8. The **Eirene** routines *check_geom_consist*, *correct_stats_input*, *couple_init*, *couple_param*, *expint*, *filepath_usr*, *find_param_fixform*, *find_tet_dim*, *find_triang_dim*, *init_eion*, *init_params*, *lax_m, lookup_adasdir_usr*, *num_background_spectra*, *prep_plotting*, *prep_strata*, *read_fixform*, *set_cell_diameter*, *set_derived_input_parameters*, *set_specific_zones*, *set_vis_add_surf_ranges*, *setup_atomic_weights*, *setup_inmp*, *setup_ispez*, *setup_spec_ind_distrib*, *setup_surface_switches*, *setup_time_surface*, and *time2surface* have been extracted to be in their own separate files. Conversely, routines *rstern*, *rtsaf*, *galpd_m*, *gaumeh*, *fi*, and *fivec* have been consolidated into the modules that use them. All the *h1rn*.f* routines have been grouped together into the *eirmod_h1rnm.f* module.
9. Obsolete photon transport features and routines *eirmod_sighe.F* and *he_emis.f* have been removed from the **Eirene** distribution. Replacement routines *sigeir.f*, *siglos.f*, and *sigpla.f* are in place.
10. Within **B2.5**, switches that are needed to define the default value of others can be obtained first by putting them in the *read_first_switches* routine.
11. Some routines that have no purpose in the unstructured code have been removed: *b2mod_nclass.F*, *b2nxpo.F*, *b2ptrg.F*, *b2sihs.F*, *b2stbc_bas.F*, *b2stbc_fb.F*, *b2stbc_spb.F*, *b2stbr_bas.F*, *b2txcv.F*, *b2usr_loads.F*, *eirflux_map.F*, *map.F*, *mapx.F*, *mapy.F*, and *profile_average.F*.
12. Whenever relevant, we now make use of the *set_identifier* API in the IMAS Data Dictionary to allow for naming aliases when filling in IDS data.
13. In **Eirene**, use the `MASTER_PATH` variable instead of the *get_solpstop()* function to specify the prefix to a file path. When opening a file, use the `EIRENE_OPENFILE` function.
14. The new `write_manifest` routine in *b2mod_ual.F90* contains a list of the potential I/O files that may be included in a SimDB manifest. Thus, when adding a file to the code workflow, this list needs to be updated accordingly.
15. Some parts of the code have not yet been ported from the structured code and are bracketed with `#ifdef WG_TODO` pragmas. Most of these are rarely-used optional features for which coding effort may be devoted if a dedicated use case is brought forward, but otherwise are considered low priority for implementation.
16. Use of the `lnblnk` intrinsic function has been replaced with other equivalent functions and should be avoided.
17. To avoid another intrinsic name clash, the *floor* single-precision function has been renamed as *sfloor*.
18. The *samax* function has been renamed *damax* to reflect that it is double-precision.